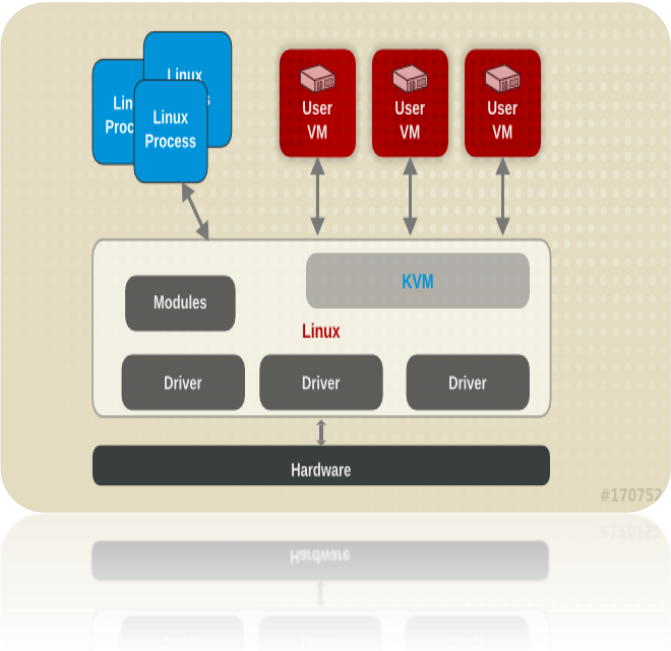


CLOUD COMPUTING COURSE

Lecture11-KVM & Fusion Compute

ABSTRACT

Lecture11 Introduces KVM and Fusion Compute Virtualization Products

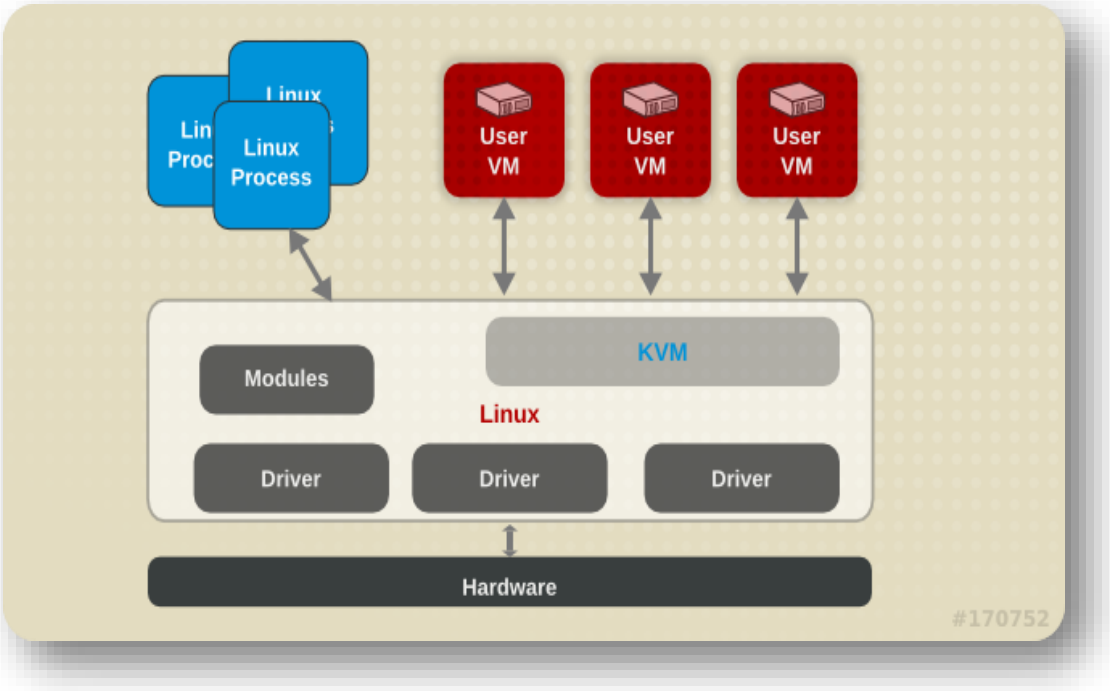
Instructor: Asma'a Khtoom

Virtualization Products

KVM Virtualization Product

Huawei virtualization products earlier than the 6.3 version are developed based on Xen. In 6.3 and later, they are developed based on Kernel-based Virtual Machine (KVM).

KVM is a **Type-II full virtualization** solution. It is a **Linux** kernel module. A physical machine with a Linux kernel module installed can function as a hypervisor, which does not affect the other applications running on the Linux OS.



After the KVM module is installed in a common Linux OS, three running modes are added:

1. **Guest Mode:** VMs, including their CPUs, memory, and disks.
2. **User Mode:** The quick emulator (QEMU) typically runs in this mode. QEMU emulates I/O requests.
3. **Kernel Mode**: In this mode, the hardware can be operated. When the guest OS executes an I/O operation or privileged instruction, a request needs to be submitted to the user mode, and then the user mode initiates a hardware operation request to the kernel mode again to operate the hardware.

A KVM architecture consists of three parts:

1. KVM kernel module:

The KVM kernel module is the core of a KVM VM. This module initializes the CPU hardware, enables the virtualization mode, runs the guest machine in the Guest mode, and supports the running of the virtual client.

- **KVM runs in the kernel space.**
- **Only provides CPU and memory virtualization.**

2. QEMU:

A VM requires other I/O devices such as NICs and hard disks besides CPUs and memory. QEMU is required to implement NIC and hard disk virtualization functions.

QEMU **was not a part of KVM kernel**. It was a universal **open-source** virtualization emulator that uses pure software to implement virtualization. The guest OS considers that it is interacting with hardware. Actually, QEMU is interacting with hardware. This means that all interactions with the hardware need to pass through QEMU. Therefore, the simulation **performance delivered by QEMU is low**.

- **QEMU runs in the user space**

3. Management tool.

In addition to virtualization of various devices, QEMU-KVM provides native tools for creating, modifying, and deleting VMs. However, **Libvirt** is the most widely used tool and API for managing KVM VMs.

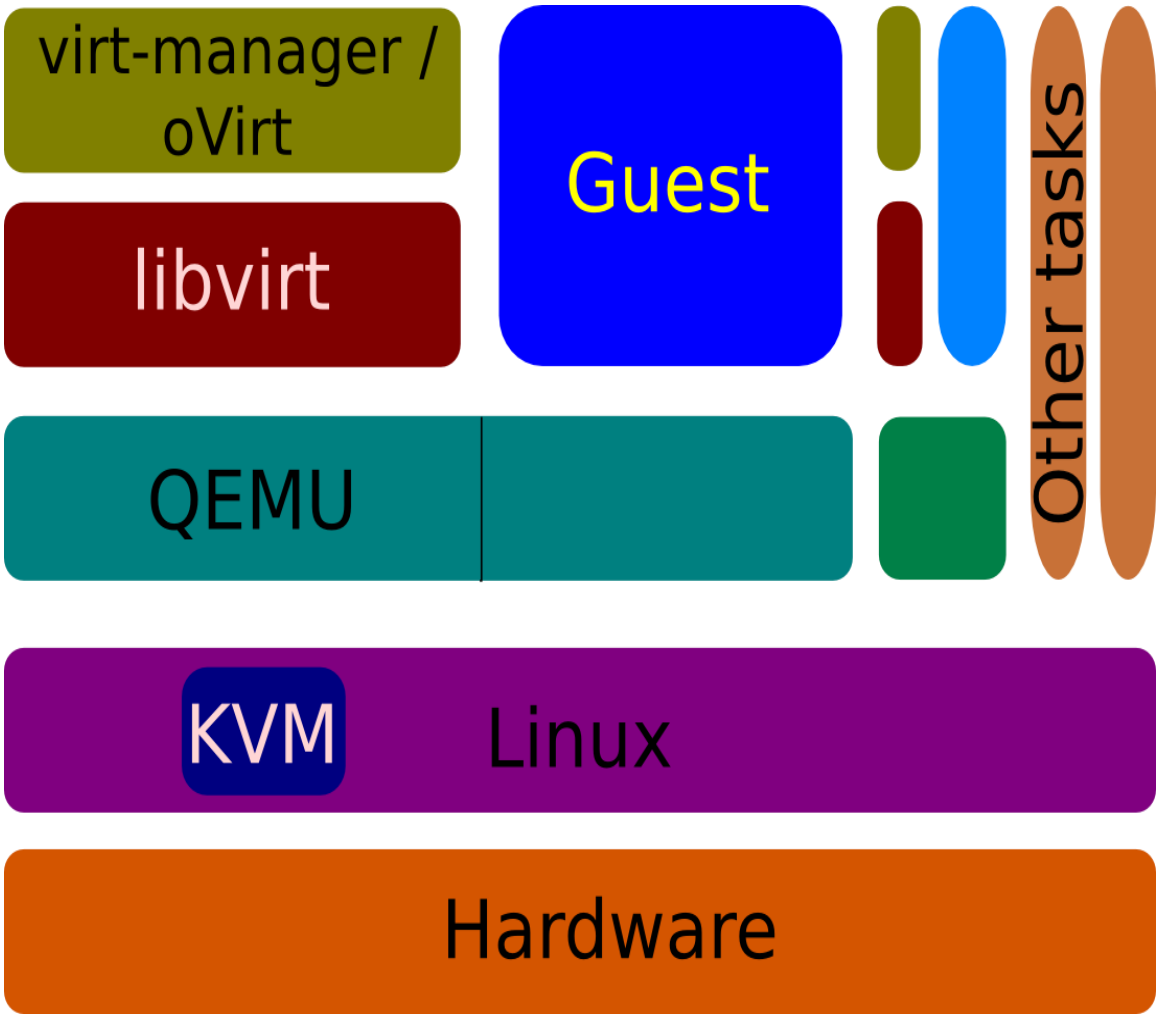
Libvirt is:

1. An **open-source** project and is a powerful management tool. It is able to manage virtualization platforms such as KVM, Xen, VMware, and Hyper-V.

2. Libvirt is an API developed using the **C language**.

3. In cloud computing, there are various hypervisors. Each hypervisor has its own management tool, and parameters are complex and difficult to use. Hypervisors **are not unified**, and there is no unified **programming interface** to manage them, which severely affects the cloud computing environment. With **Libvirt**, it can connect to various hypervisors, such as KVM and Xen, and provide APIs in various languages.

4. Libvirt serves as the **middle layer** between the management tool and hypervisor and is completely transparent to upper-layer users.


virt-manager / oVirt
libvirt
Guest
QEMU
Other tasks
KVM
Linux
Hardware

FusionCompute Virtualization Product

Virtualizes server, storage, and network resources, providing elastic resource pools for automatic resource scheduling and management. It consists of Compute Node Agent (CNA), a virtualization engine, and Virtual Resource Management (VRM).

It is based on **the Linux OS.**

Benefit of FusionCompute

Following benefits for customers:

1. Improves infrastructure resource utilization data centers.
2. Significantly accelerates service rollout.
3. Substantially reduces power consumption in data centers.
4. Provides rapid automatic fault recovery for services, decreases data center costs, and increases system runtime by leveraging high availability and powerful restoration capabilities of virtualized infrastructure.

FusionCompute Parts

FusionCompute consists of two parts:

1. **Computing Node Agent (CNA)**

   CNA manages VMs and resources on the local node (the server), it provides the virtualization function, and it based on the Linux OS.

CNA provides the following functions:

1. Provides the virtual computing function.
2. Manages the VMs running on compute nodes.
3. Manages compute, storage, and network resources on compute nodes.

2. Virtual Resource Manager (VRM):

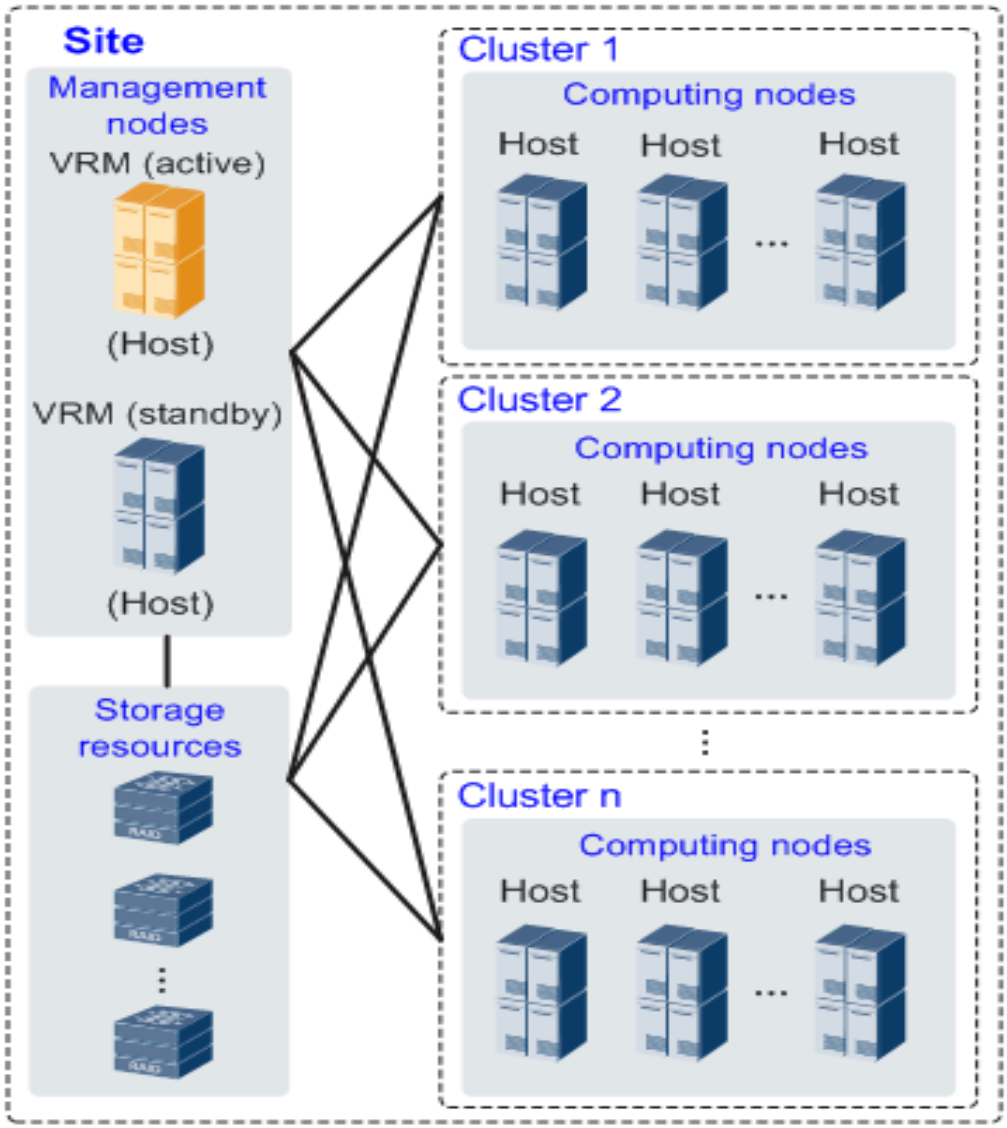
VRM is the resource scheduling and management software.

VRM manages clusters or resources in the resource pool. It functions as **the management tool** of KVM.

Administrators and common users can manage and use FusionCompute on the GUI-based portal of VRM.

VRM provides the following functions:

1. Manages block storage resources in a cluster.
2. Manages network resources, such as IP addresses and virtual local area network (VLAN) IDs, in a cluster and allocates IP addresses to VMs.
3. Manages the lifecycle of VMs in a cluster, distributes, and migrates VMs across compute nodes.
4. Dynamically scales resources in a cluster.
5. Implements centralized management of virtual resources and user data and provides elastic computing, storage, and IP address services.


Site
Management nodes
VRM (active)
(Host)
VRM (standby)
(Host)
Storage resources
Cluster 1
Computing nodes
Host
Host
Host
Cluster 2
Computing nodes
Host
Host
Host
Cluster n
Computing nodes
Host
Host
Host